

Chapter 4

Orbital Stability and Transverse Linearization

Stability is not the absence of motion.
It is the faithfulness of motion.

4.1 The Problem with Lyapunov

The classical Lyapunov framework asks a simple question: *does the state converge to an equilibrium?* We demonstrated in Chapter 1 that this is the wrong question for systems whose purpose is to move. But the problem goes deeper than philosophical framing—Lyapunov stability is *structurally incapable* of analyzing trajectory-tracking systems.

Consider a system perfectly tracking a reference trajectory $\gamma^*(t)$. The state $\mathbf{x}(t)$ matches $\gamma^*(t)$ at every instant. Now apply a small timing perturbation: the system executes the same motion but shifted by ε seconds. The perturbed state is $\tilde{\mathbf{x}}(t) = \gamma^*(t - \varepsilon)$, and the time-indexed error is

$$\mathbf{e}(t) = \tilde{\mathbf{x}}(t) - \gamma^*(t) = \gamma^*(t - \varepsilon) - \gamma^*(t) \approx -\varepsilon \dot{\gamma}^*(t). \quad (4.1)$$

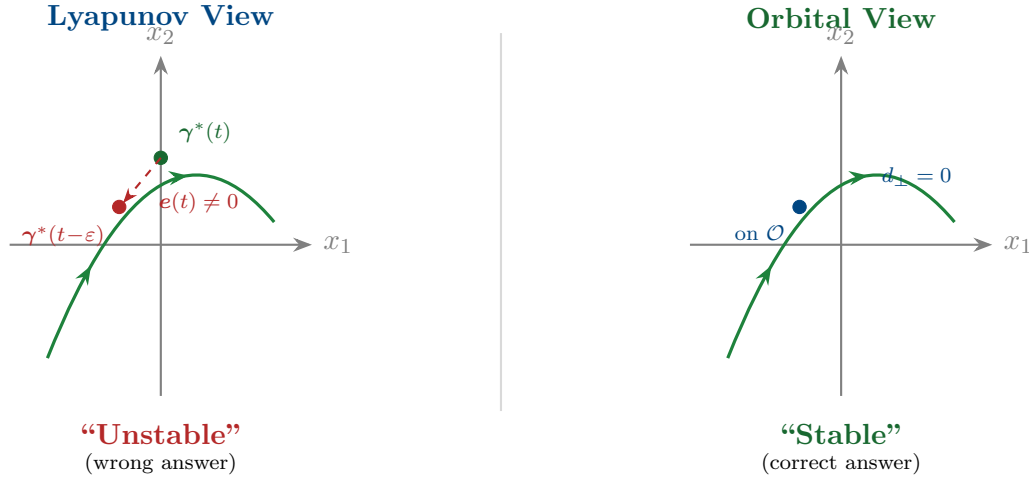
This error does *not* decay to zero—it persists indefinitely, oscillating with a magnitude proportional to $\varepsilon v(t)$ where v is the speed. By Lyapunov’s definition, the trajectory is *unstable*. But the system is executing the correct motion perfectly! The only “error” is that it is slightly ahead of or behind schedule.

Key Idea 4.1. The Fundamental Inadequacy of Point Stability

Lyapunov stability conflates two independent phenomena:

- (i) **Transverse deviations:** the system drifts *away from* the trajectory path. This is a genuine control failure.
- (ii) **Tangential deviations:** the system is ahead of or behind schedule *along* the trajectory path. This is a timing difference, not a tracking failure.

Lyapunov stability rejects both. Orbital stability accepts tangential deviations and rejects only transverse ones. For systems whose purpose is motion, orbital stability is the correct notion.



4.2 Orbital Stability: Formal Definitions

We now define orbital stability rigorously. The key object is the **orbit**—the image of the trajectory as a set, stripped of all timing information.

Definition 4.1. Orbit

Given a trajectory $\gamma^* : [0, T] \rightarrow \mathcal{X}$, the **orbit** is the set

$$\mathcal{O} = \{\gamma^*(t) : t \in [0, T]\} \subset \mathcal{X}. \quad (4.2)$$

The orbit is a one-dimensional submanifold of $\mathcal{X}$ (a curve without parameterization). For periodic trajectories with period T , $\gamma^*(0) = \gamma^*(T)$ and $\mathcal{O}$ is a closed curve (a loop).

Definition 4.2. Orbital Stability

A trajectory $\gamma^*(t)$ (or its orbit $\mathcal{O}$) is **orbitally stable** if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d(\mathbf{x}(0), \mathcal{O}) < \delta \implies d(\mathbf{x}(t), \mathcal{O}) < \varepsilon \quad \forall t \geq 0, \quad (4.3)$$

where $d(\mathbf{x}, \mathcal{O}) = \inf_{p \in \mathcal{O}} \|\mathbf{x} - p\|$ is the distance from the state to the orbit.

It is **asymptotically orbitally stable** if additionally $d(\mathbf{x}(t), \mathcal{O}) \rightarrow 0$ as $t \rightarrow \infty$.

Critically, orbital stability does not require $\|\mathbf{x}(t) - \gamma^*(t)\| \rightarrow 0$. The system may drift along the orbit (ahead or behind schedule) while remaining perfectly stable. Only deviations *away from* the orbit matter.

Definition 4.3. Orbital Stability with Asymptotic Phase

A trajectory is orbitally stable **with asymptotic phase** if it is asymptotically orbitally stable and additionally there exists a phase shift θ^* such that

$$\|\mathbf{x}(t) - \boldsymbol{\gamma}^*(t + \theta^*)\| \rightarrow 0 \quad \text{as } t \rightarrow \infty. \quad (4.4)$$

This means the system not only returns to the orbit but eventually “locks in” to a definite timing, shifted by θ^* from the nominal schedule. This is the strongest practical form of trajectory stability.

Example 4.1. Limit Cycles: The Prototype

The Van der Pol oscillator $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$ possesses a **limit cycle**—a periodic orbit that is asymptotically orbitally stable with asymptotic phase. All nearby trajectories spiral toward the limit cycle and eventually synchronize to a definite phase. The limit cycle is the simplest example of a system whose purpose is to move: the “target” is not a point but an orbit, and stability means the orbit attracts nearby trajectories. Walking, running, and heartbeats are all limit cycles. The golf swing is a transient trajectory (not periodic), but the same stability concepts apply over the finite interval $[0, T]$.

4.3 The Transverse–Tangential Decomposition

To analyze orbital stability, we must decompose the dynamics into components along and across the orbit. This is the *transverse–tangential decomposition*, the central technical tool of this chapter.

4.3.1 Projection onto the Orbit

Definition 4.4. Nearest-Point Projection

Given a state $\mathbf{x}$ sufficiently close to the orbit $\mathcal{O}$, the **nearest-point projection** is

$$\pi(\mathbf{x}) = \arg \min_{p \in \mathcal{O}} \|\mathbf{x} - p\|, \quad (4.5)$$

and the associated **projected phase** is the unique $s^*(\mathbf{x})$ satisfying $\pi(\mathbf{x}) = \boldsymbol{\gamma}^*(s^*)$. The projection is well-defined and smooth in a tubular neighborhood of $\mathcal{O}$ (as long as $\mathbf{x}$ is closer to $\mathcal{O}$ than the minimum radius of curvature of the orbit).

4.3.2 The Moving Hyperplane Decomposition

At each point $\boldsymbol{\gamma}^*(s)$ on the orbit, the state space $\mathbb{R}^n$ decomposes into a one-dimensional tangential direction and an $(n-1)$ -dimensional transverse hyperplane:

Definition 4.5. Transverse Hyperplane

At the point $\gamma^*(s)$ on the orbit, define:

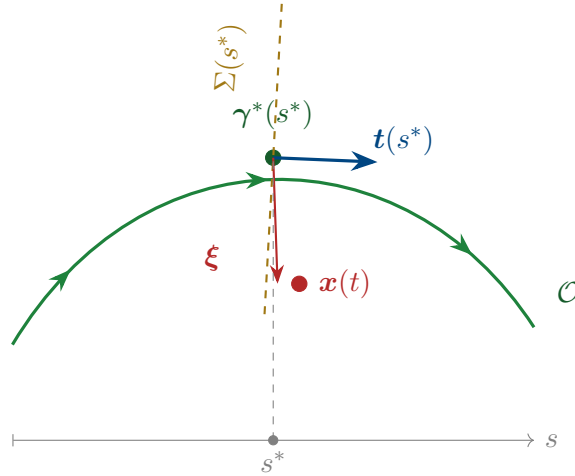
$$\text{Tangential direction: } \mathbf{t}(s) = \frac{\dot{\gamma}^*(s)}{\|\dot{\gamma}^*(s)\|}, \quad (4.6)$$

$$\text{Transverse hyperplane: } \Sigma(s) = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^\top \mathbf{t}(s) = 0\}. \quad (4.7)$$

A state $\mathbf{x}$ near $\gamma^*(s)$ decomposes as

$$\mathbf{x} = \gamma^*(s^*) + \boldsymbol{\xi}, \quad \boldsymbol{\xi} \in \Sigma(s^*), \quad (4.8)$$

where s^* is the projected phase and $\boldsymbol{\xi}$ is the **transverse deviation**—the component of the error perpendicular to the orbit. Orbital stability is equivalent to the stability of the origin $\boldsymbol{\xi} = \mathbf{0}$ in the transverse dynamics.



4.4 Transverse Linearization

The power of the transverse decomposition is that it reduces orbital stability to the stability of a *time-varying linear system*—the transverse linearization.

4.4.1 Deriving the Transverse Dynamics

Consider the nonlinear system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$ with a reference trajectory $\gamma^*(t)$ satisfying $\dot{\gamma}^* = \mathbf{f}(\gamma^*, \mathbf{u}^*(t))$. Using the decomposition $\mathbf{x} = \gamma^*(s^*) + \boldsymbol{\xi}$ and linearizing about $\boldsymbol{\xi} = \mathbf{0}$:

Theorem 4.1 (Transverse Linearization). *The linearized transverse dynamics about the orbit $\mathcal{O}$ are*

$$\dot{\boldsymbol{\xi}} = \mathbf{A}_\perp(s) \boldsymbol{\xi} + \mathbf{B}_\perp(s) \delta \mathbf{u}, \quad (4.9)$$

where $\delta \mathbf{u} = \mathbf{u} - \mathbf{u}^*$ is the feedback perturbation, and

$$\mathbf{A}_\perp(s) = \mathbf{P}(s) \left[\left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\gamma^*, \mathbf{u}^*} - \frac{\dot{\gamma}^* \otimes (\nabla_{\mathbf{x}} s^*)^\top}{\|\dot{\gamma}^*\|^2} \cdot \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\gamma^*, \mathbf{u}^*} \right] \mathbf{P}(s), \quad (4.10)$$

$$\mathbf{B}_\perp(s) = \mathbf{P}(s) \left. \frac{\partial \mathbf{f}}{\partial \mathbf{u}} \right|_{\gamma^*, \mathbf{u}^*}, \quad (4.11)$$

and $\mathbf{P}(s) = \mathbf{I} - \mathbf{t}(s)\mathbf{t}(s)^\top$ is the orthogonal projector onto $\Sigma(s)$.

Remark 4.1. Understanding the Transverse $\mathbf{A}_\perp$

Equation (4.10) has a clear structure:

- The first term $\mathbf{P} \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \mathbf{P}$ is the standard Jacobian projected onto the transverse space. This captures how the system dynamics affect cross-trajectory deviations.
- The second term subtracts the component of the Jacobian that acts along the tangent direction—this is exactly the tangential dynamics that orbital stability “forgets.”

The resulting $\mathbf{A}_\perp(s)$ has dimension $(n-1) \times (n-1)$, one less than the full state Jacobian. The removed eigenvalue corresponds to motion along the orbit—the tangential direction in which we are *not* demanding stability.

Principle 4.1. The Transverse Reduction Principle

The reference trajectory $\gamma^*(t)$ is asymptotically orbitally stable if and only if the origin of the transverse linearization (4.9) (with $\delta \mathbf{u} = \mathbf{0}$) is asymptotically stable. This reduces the orbital stability question from a nonlinear problem on the full n -dimensional state space to a *linear* problem on the $(n-1)$ -dimensional transverse space. For controller design: if we can stabilize the linear system (4.9), we achieve orbital stability of the nonlinear system. All the tools of linear time-varying (LTV) control theory apply.

4.4.2 The Transverse Jacobian: A Practical Formula

For implementation, the key quantities are computed as follows. Let $\mathbf{J}(t) = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \right|_{\gamma^*, \mathbf{u}^*}$ be the Jacobian along the reference. Choose any orthonormal basis $\{\mathbf{n}_1(s), \dots, \mathbf{n}_{n-1}(s)\}$ for $\Sigma(s)$ —the Frenet–Serret frame from Chapter 2 provides a natural choice. Collect these into the matrix $\mathbf{N}(s) = [\mathbf{n}_1 \mid \dots \mid \mathbf{n}_{n-1}] \in \mathbb{R}^{n \times (n-1)}$. Then:

$$\mathbf{A}_\perp(s) = \mathbf{N}(s)^\top \left[\mathbf{J}(s) - \frac{\mathbf{f}^* \cdot \mathbf{J}(s)^\top \mathbf{t}(s)}{\|\mathbf{f}^*\|^2} \mathbf{f}^* \mathbf{t}(s)^\top - \dot{\mathbf{N}}(s) \mathbf{N}(s)^\top \right] \mathbf{N}(s), \quad (4.12)$$

where $\mathbf{f}^* = \mathbf{f}(\gamma^*, \mathbf{u}^*)$ and $\dot{\mathbf{N}}$ accounts for the rotation of the transverse frame along the orbit (the Frenet–Serret equations).

Example 4.2. Transverse Linearization of a Planar System

Consider the system $\dot{x}_1 = x_2$, $\dot{x}_2 = -x_1 + u$ tracking the circular orbit $\gamma^*(t) = (\cos t, -\sin t)$ with $u^* = 0$.

The Jacobian is $\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, the tangent vector is $\mathbf{t} = (-\sin t, -\cos t)^\top / 1 = (-\sin t, -\cos t)^\top$, and the normal is $\mathbf{n} = (-\cos t, \sin t)^\top$.

Computing $\mathbf{A}_\perp = \mathbf{n}^\top [\mathbf{J} - \text{tangential terms}] \mathbf{n}$, we find:

$$\mathbf{A}_\perp(t) = 0, \quad \mathbf{B}_\perp(t) = \mathbf{n}^\top \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \sin t. \quad (4.13)$$

The open-loop transverse dynamics are *neutrally stable* ($\mathbf{A}_\perp = 0$). Feedback $\delta u = -k \xi$ gives $\dot{\xi} = -k \sin^2 t \xi$ (after averaging), which is stable for $k > 0$. The orbit is stabilizable.

4.5 Floquet Theory: Stability of Periodic Orbits

For periodic trajectories (limit cycles, walking gaits, repetitive manufacturing), the transverse linearization is a *periodic* linear system. Floquet theory provides the definitive stability analysis tool.

4.5.1 The State Transition Matrix

Definition 4.6. State Transition Matrix

For the linear time-varying system $\dot{\xi} = \mathbf{A}_\perp(t) \xi$, the **state transition matrix** $\Phi(t, t_0)$ satisfies

$$\dot{\Phi}(t, t_0) = \mathbf{A}_\perp(t) \Phi(t, t_0), \quad \Phi(t_0, t_0) = \mathbf{I}. \quad (4.14)$$

The solution is $\xi(t) = \Phi(t, t_0) \xi(t_0)$.

4.5.2 The Monodromy Matrix

Definition 4.7. Monodromy Matrix

For a periodic orbit with period T , the **monodromy matrix** is the state transition matrix over one complete period:

$$\mathbf{M} = \Phi(T, 0). \quad (4.15)$$

The eigenvalues $\mu_1, \dots, \mu_{n-1}$ of $\mathbf{M}$ are the **Floquet multipliers** (also called **characteristic multipliers**).

Theorem 4.2 (Floquet Stability Criterion). *A periodic orbit is:*

- (i) **Orbitally stable** if all Floquet multipliers satisfy $|\mu_k| \leq 1$, with multipliers on the unit circle being semisimple.
- (ii) **Asymptotically orbitally stable** if all Floquet multipliers satisfy $|\mu_k| < 1$ (all strictly inside the unit circle).
- (iii) **Orbitally unstable** if any Floquet multiplier satisfies $|\mu_k| > 1$.

Remark 4.2. The Missing Multiplier

If we had computed the monodromy matrix for the *full* n -dimensional system (not the transverse reduction), there would be n Floquet multipliers, with one of them exactly equal to $+1$. This “trivial” multiplier corresponds to the tangential direction—perturbations along the orbit neither grow nor decay. The transverse reduction removes this trivial multiplier, leaving only the $n - 1$ multipliers that matter for orbital stability.

This is why the transverse reduction is essential: without it, the monodromy matrix *always* has an eigenvalue at $+1$, making the orbit appear only marginally stable even when it is asymptotically orbitally stable.

Example 4.3. Floquet Analysis of a Walking Gait

Consider a simplified compass-gait biped with 4-dimensional state space $(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2)$. The walking gait is a periodic orbit with period T (one stride). The transverse linearization has dimension $n - 1 = 3$.

After numerically integrating $\Phi(T, 0)$ for the transverse dynamics, suppose we find the monodromy matrix has eigenvalues $\mu_1 = 0.7$, $\mu_2 = 0.4$, $\mu_3 = -0.3$. Since all $|\mu_k| < 1$, the gait is asymptotically orbitally stable.

The physical interpretation: $\mu_1 = 0.7$ means that after one stride, transverse deviations in the first mode are reduced to 70% of their initial value. After 10 strides, they are reduced to $0.7^{10} \approx 3\%$. The negative multiplier $\mu_3 = -0.3$ indicates an alternating (flip-flop) convergence pattern in the third mode.

For the golf swing (a non-periodic motion), we cannot use the monodromy matrix directly, but the state transition matrix $\Phi(T, 0)$ plays an analogous role—its singular values measure how much transverse deviations amplify or attenuate over the course of the motion.

4.6 Moving Poincaré Sections

A complementary tool for analyzing orbital stability is the **Poincaré section**—a codimension-1 surface that the orbit crosses transversally. While Floquet theory gives continuous-time information, Poincaré sections give *discrete-time* (strobed) information.

Definition 4.8. Poincaré Section

A **Poincaré section** for an orbit $\mathcal{O}$ is a smooth $(n-1)$ -dimensional surface $\mathcal{S} \subset \mathcal{X}$ such that:

- (i) $\mathcal{O}$ intersects $\mathcal{S}$ transversally (not tangentially).
- (ii) In a neighborhood of the intersection, every orbit near $\mathcal{O}$ also crosses $\mathcal{S}$.

The **Poincaré return map** $P : \mathcal{S} \rightarrow \mathcal{S}$ sends each point on $\mathcal{S}$ to the next intersection with $\mathcal{S}$ under the flow.

For periodic orbits, the orbit crosses $\mathcal{S}$ once per period, and the Poincaré return map has a fixed point at the intersection $\mathbf{x}^* = \mathcal{O} \cap \mathcal{S}$. Orbital stability of $\mathcal{O}$ is equivalent to stability of this fixed point under the return map.

Theorem 4.3 (Poincaré–Floquet Connection). *The Jacobian of the Poincaré return map at the fixed point equals the transverse monodromy matrix:*

$$DP(\mathbf{x}^*) = M_{\perp}. \quad (4.16)$$

The Floquet multipliers and the eigenvalues of the Poincaré map are identical.

4.6.1 Moving Poincaré Sections for Non-Periodic Trajectories

The golf swing is not periodic—it is a *finite-time* trajectory. The classical Poincaré section (which relies on periodicity) does not directly apply. However, we can generalize to **moving Poincaré sections**: a continuous family of transverse hyperplanes, one at each point of the orbit.

Definition 4.9. Moving Poincaré Section

A **moving Poincaré section** along a trajectory $\gamma^*(s)$, $s \in [0, L]$, is the family of hyperplanes

$$\mathcal{S}(s) = \gamma^*(s) + \Sigma(s), \quad (4.17)$$

where $\Sigma(s)$ is the transverse hyperplane at s . The **section-to-section map** from $\mathcal{S}(s_1)$ to $\mathcal{S}(s_2)$ is defined by following the flow from s_1 to s_2 :

$$P_{s_1 \rightarrow s_2} : \mathcal{S}(s_1) \rightarrow \mathcal{S}(s_2), \quad \xi(s_1) \mapsto \xi(s_2). \quad (4.18)$$

The linearized section-to-section map is the state transition matrix $\Phi_{\perp}(s_2, s_1)$ of the transverse dynamics.

For the golf swing, this gives us a powerful analysis tool:

Principle 4.2. Finite-Time Orbital Stability

For a finite-time trajectory $\gamma^* : [0, T] \rightarrow \mathcal{X}$, define the **transverse amplification ratio** from phase s_1 to s_2 :

$$\rho(s_1, s_2) = \|\Phi_\perp(s_2, s_1)\|_2 = \sigma_{\max}(\Phi_\perp(s_2, s_1)), \quad (4.19)$$

where $\sigma_{\max}$ is the largest singular value. If $\rho(s_1, s_2) < 1$, then transverse deviations shrink between s_1 and s_2 . If $\rho(s_1, s_2) > 1$, they grow.

For trajectory design, we require:

$$\rho(0, s_{\text{impact}}) \ll 1 \quad (\text{deviations shrink toward impact}). \quad (4.20)$$

This is a precise formulation of the funnel concept from Chapter 1: the tube narrows because the transverse dynamics are contracting.

Example 4.4. Amplification Profile of the Golf Swing

Consider the 8-dimensional state space of a 4-DOF golf swing model. The transverse dynamics have dimension 7. The transverse state transition matrix $\Phi_\perp(s, 0)$ maps initial deviations to deviations at phase s .

Numerical computation reveals the amplification ratio $\rho(0, s)$ as a function of s :

- $s \in [0, 0.3]$ (backswing): ρ is approximately 1—deviations neither grow nor shrink. The motion is kinematically slow and nearly linear.
- $s \in [0.3, 0.5]$ (transition): ρ *increases* past 1. This is the unstable phase—the high curvature and speed change amplify deviations. Without feedback, errors grow here.
- $s \in [0.5, 0.7]$ (late downswing): ρ *decreases*. The passive dynamics of the double-pendulum release are self-stabilizing in the transverse directions. The wrist release “funnels” the club toward the correct impact state.
- $s = 0.75$ (impact): $\rho(0, 0.75) < 1$. The net effect over the entire swing is transverse contraction—deviations are smaller at impact than at address.

This profile explains why skilled golfers are accurate despite the extreme nonlinearity of the swing: the trajectory is designed so that its *passive transverse dynamics* are contracting near impact, creating a natural funnel that corrects errors without active feedback.

4.7 Controller Design via Transverse Stabilization

We now use the transverse linearization to design tracking controllers. The approach is systematic: stabilize the transverse linear system (4.9) using standard LTV techniques, then

map the result back to the nonlinear system.

4.7.1 Time-Varying LQR for Transverse Stabilization

Principle 4.3. Transverse LQR

The trajectory tracking problem reduces to solving the *differential Riccati equation* for the transverse dynamics:

$$-\dot{\mathbf{S}}(t) = \mathbf{A}_\perp^\top \mathbf{S} + \mathbf{S} \mathbf{A}_\perp - \mathbf{S} \mathbf{B}_\perp \mathbf{R}^{-1} \mathbf{B}_\perp^\top \mathbf{S} + \mathbf{Q}(t), \quad (4.21)$$

with terminal condition $\mathbf{S}(T) = \mathbf{S}_T$. The optimal transverse feedback gain is

$$\mathbf{K}_\perp(t) = \mathbf{R}^{-1} \mathbf{B}_\perp(t)^\top \mathbf{S}(t). \quad (4.22)$$

The full control law is

$$\mathbf{u}(t) = \underbrace{\mathbf{u}^*(t)}_{\text{feedforward}} + \underbrace{\mathbf{N}(s^*) \mathbf{K}_\perp(t) \mathbf{N}(s^*)^\top (\mathbf{x} - \boldsymbol{\gamma}^*(s^*))}_{\text{transverse feedback}}, \quad (4.23)$$

where $\mathbf{N}(s^*)$ maps between the transverse coordinates $\boldsymbol{\xi}$ and the full state coordinates.

The weighting matrices $\mathbf{Q}(t)$ and $\mathbf{R}$ have clear physical interpretations:

- $\mathbf{Q}(t)$ penalizes transverse deviation. Making $\mathbf{Q}$ large near impact tightens the funnel where precision matters.
- $\mathbf{R}$ penalizes control effort. Making $\mathbf{R}$ small allows aggressive correction but may demand unrealistic actuator authority.
- The *phase-dependent* nature of $\mathbf{Q}(t)$ is essential: we want tight tracking near impact and loose tracking during the backswing. This is the time-varying tube from Chapter 1, now realized through the Riccati equation.

4.7.2 Phase-Varying Gains and the Funnel Connection

The terminal-value Riccati equation (4.21) naturally produces gains that tighten as the terminal time approaches. This is exactly the funnel behavior we designed geometrically in Chapter 1:

Theorem 4.4 (Riccati Funnel). *Let $\mathbf{S}(t)$ be the solution of (4.21) with terminal penalty $\mathbf{S}_T \succ 0$. The ellipsoidal set*

$$\mathcal{E}(t) = \{\boldsymbol{\xi} \in \Sigma(s) : \boldsymbol{\xi}^\top \mathbf{S}(t) \boldsymbol{\xi} \leq c\} \quad (4.24)$$

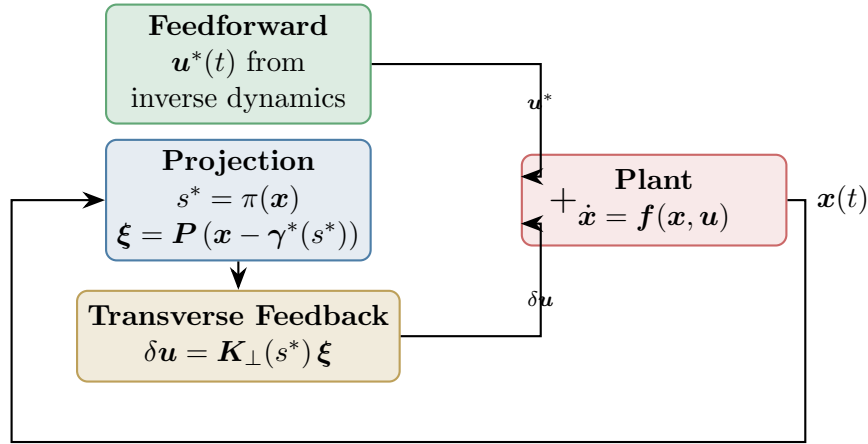
defines a time-varying tube whose width (in the direction of eigenvector $\mathbf{v}_k$ of $\mathbf{S}$) is proportional to $1/\sqrt{\lambda_k(t)}$, where $\lambda_k(t)$ is the corresponding eigenvalue of $\mathbf{S}(t)$.

As $t \rightarrow T$ and $\mathbf{S}(t) \rightarrow \mathbf{S}_T$, the tube tightens. The closed-loop transverse dynamics ensure that any trajectory starting inside the tube at $t = 0$ remains inside for all $t \in [0, T]$, provided the control authority is sufficient.

This is the formal connection between the geometric funnel of Chapter 1 and the algebraic Riccati equation: **the funnel is the level set of the Riccati cost-to-go.**

4.7.3 The Complete Tracking Controller Architecture

Assembling all the pieces, the trajectory tracking controller has three layers:



Layer 1. Feedforward $u^*(t)$: computed offline from inverse dynamics, provides $\sim 90\%$ of the total control effort.

Layer 2. Projection: measures the current state $\mathbf{x}(t)$, projects onto the orbit to find s^* , and computes the transverse deviation ξ .

Layer 3. Transverse feedback $\delta u = K_\perp(s^*) \xi$: stabilizes the transverse dynamics using phase-varying gains from the Riccati equation.

Remark 4.3. Why Not Just Use Standard LQR?

One could linearize the *full* dynamics about $\gamma^*(t)$ and apply standard time-varying LQR to the n -dimensional error $\mathbf{e}(t) = \mathbf{x}(t) - \gamma^*(t)$. This works, but:

- (a) It “wastes” control effort trying to correct tangential deviations (timing errors), which are benign.
- (b) It fights the inherent marginal stability in the tangential direction, leading to unnecessarily high gains.
- (c) It does not produce the correct funnel shape—it penalizes all errors equally, including timing errors.

The transverse approach avoids all three problems. It is geometrically correct, computationally smaller (dimension $n - 1$ vs. n), and physically meaningful.

4.8 Robustness of Orbital Stability

A practical controller must be robust to model uncertainty and external disturbances. The transverse framework provides natural robustness margins.

Definition 4.10. Transverse Gain and Phase Margins

For the closed-loop transverse system $\dot{\xi} = [A_{\perp} - B_{\perp}K_{\perp}]\xi$, the **transverse gain margin** and **transverse phase margin** are the gain and phase margins of the transverse loop transfer function. These are the trajectory-tracking analogues of the classical setpoint stability margins.

Proposition 4.1 (Robustness of Transverse LQR). *If the transverse LQR is designed with $R = rI$, then the closed-loop transverse system has guaranteed gain margin $[1/2, \infty)$ and phase margin $\pm 60^\circ$, inheriting the classical LQR robustness guarantees in each transverse channel.*

For the golf swing, robustness translates directly to *repeatability*: a swing with high transverse robustness margins produces consistent results even when the golfer’s motor commands are noisy.

4.9 Summary

Concept	Significance
Orbital stability	The correct stability notion for moving systems: only cross-orbit deviations matter
Transverse linearization	Reduces orbital stability to $(n-1)$ -dim LTV stability; removes the tangential “nuisance” direction
Floquet multipliers	Eigenvalues of the transverse monodromy matrix; determine stability of periodic orbits
Moving Poincaré sections	Extend Poincaré analysis to non-periodic trajectories; connect to the state transition matrix
Amplification ratio ρ	Maximum singular value of $\Phi_{\perp}$; quantifies transverse error growth or contraction
Transverse LQR	Phase-varying feedback from the Riccati equation; the optimal funnel controller
Riccati funnel	The LQR cost-to-go defines the funnel boundary; connects geometric and algebraic views

This chapter has provided the mathematical backbone for everything that follows. The transverse linearization converts the nonlinear trajectory-tracking problem into a linear, lower-dimensional problem that inherits all the powerful tools of LTV control theory. The Floquet/Poincaré framework provides both analytical insight and numerical algorithms for assessing stability.

The key insight for the golf swing is that orbital stability—not Lyapunov stability—is the correct framework. The swing is designed so that its passive transverse dynamics create a natural funnel toward impact, and the transverse LQR controller tightens this funnel using phase-varying gains that match the demands of each phase of the motion.

In Chapter 5, we will see how underactuation constrains the transverse dynamics, creating geometric structure that can be *exploited* rather than merely tolerated.

4.10 Exercises

- 4.1. (Conceptual.)** Explain why a pendulum swinging on its limit cycle (pumped by a periodic torque) is orbitally stable but *not* Lyapunov stable. Sketch the phase portrait and identify the tangential and transverse directions.
- 4.2. (Computation.)** For the Van der Pol oscillator $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$ with $\mu = 1$:
- (a) Numerically find the limit cycle by simulating from $(x, \dot{x}) = (3, 0)$ until transient effects decay.
 - (b) Compute the transverse linearization along the limit cycle.
 - (c) Compute the monodromy matrix and Floquet multipliers numerically.
 - (d) Verify that one multiplier of the full (2D) system is exactly 1.
- 4.3. (Transverse linearization.)** For the system $\dot{x}_1 = x_2 + u$, $\dot{x}_2 = -x_1^3$, consider the trajectory $\gamma^*(t) = (\cos t, -\sin t)$ with appropriate u^* .
- (a) Find $u^*(t)$ such that γ^* is a solution.
 - (b) Compute the full Jacobian $\mathbf{J}(t)$ along γ^* .
 - (c) Compute the transverse Jacobian $\mathbf{A}_\perp(t)$.
 - (d) Is the orbit stable open-loop? Design stabilizing transverse feedback.
- 4.4. (Numerical project: Transverse LQR.)** For the double pendulum from Exercise 3.4:
- (a) Compute a reference swing trajectory using direct collocation (or load from a data file).
 - (b) Compute the transverse linearization along the trajectory.
 - (c) Solve the differential Riccati equation with $\mathbf{Q}(t)$ that increases 10-fold near the terminal phase (“impact”).

- (d) Simulate the closed-loop system with random initial perturbations and plot the transverse deviation $\|\boldsymbol{\xi}(t)\|$ over time.
 - (e) Compute the amplification ratio profile $\rho(0, s)$ and identify the “natural funnel” phase.
- 4.5. (Design.)** A walking robot with period-1 gait has Floquet multipliers $\mu_1 = 0.9$, $\mu_2 = 1.1$, $\mu_3 = 0.5$.
- (a) Is the gait orbitally stable? Which mode is problematic?
 - (b) Design a transverse feedback controller that places the unstable multiplier μ_2 at 0.6. What gain is required?
 - (c) After stabilization, what is the worst-case convergence rate per stride?
 - (d) How many strides does it take to reduce a 10% transverse deviation to 1%?
- 4.6. (Philosophical.)** Compare the transverse LQR approach (this chapter) with the standard trajectory-tracking LQR (linearize about $\boldsymbol{\gamma}^*(t)$, regulate $\mathbf{e} = \mathbf{x} - \boldsymbol{\gamma}^*(t)$ to zero).
- (a) Under what conditions do the two approaches give identical feedback gains? (Hint: consider straight-line trajectories.)
 - (b) Construct a numerical example where standard LQR uses significantly more control effort than transverse LQR.
 - (c) Discuss which approach is preferable for the golf swing and why.

*The pendulum does not care if you are watching at noon or midnight.
Its orbit knows no clock.*

*Stability is not arrival. It is the promise
that the motion, once begun, will keep its shape—
even when the world pushes back.*